

Data Link Layer

Module 07

Duration: 22:39

Video: <https://www.youtube.com/watch?v=u9QZrz8pG6A>

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0:00 sing from Department of computer science Punjabi University patiala today's lecture is on data link layer in this

0:07 lecture I'll explain you the working of second layer of OSI reference model that

0:13 is data link layer now let's start in the Osa reference model the layer which

0:20 is directly on top of physical layer is the data link layer or also called dll the dll is also named as Layer Two or of

0:30 the OSI model data link layer is one of the most complex layers and has difficult functionalities and

0:37 responsibilities this layer hides the particulars of underlying hardware and depicts itself to Upper layers as the

0:46 channel to communicate data link lir works between two communicating hosts which are connected directly in some

0:54 sense this direct connection may possibly be broadcast or point to point systems on broadcast network are

1:02 supposed to be on same link the work of data link layer becomes more difficult when it is dealing with several hosts on

1:11 single collision domain data link layer converts the incoming data stream into signals bit by bit and to send it over

1:19 the underlying Hardware at the receiving end data link lay collects the data from underlying Hardware which is in the form

1:27 of electrical signals build them in an identifiable frame format and pass it over to upper layer data link layer has

1:35 two sub layers logical link control and media Access Control logical link control also called

1:44 LLC this sublayer is needed in the case of Point topoint links only it deals with protocols flow control and error

1:52 control media access control the Mac sublayer is in charge of deciding which

1:59 user get to use the broadcast medium at any particular time dll protocols can be implemented either in Hardware or

2:08 software they can be implemented in either a computer's main CPU or in special
purpose Hardware known as

2:16 network adapter or network interface card n i a common example of the later case
is

2:24 an Ethernet Nic data link lay performs many functions on behalf of upper layer the
first function is framing data link

2:33 layer receives packets from Network layer and encapsulates them into frames the
next function is addressing the data

2:41 link layer provides layer to Hardware addressing system Hardware address is
supposed to be unique and is encoded

2:50 into Hardware at the time of manufacturing the third function is error control
sometimes during

2:57 transmission signals May encounter problem and the bit get flipped so the
detection of errors and recovery of

3:05 original data is done by this layer the next function is flow control stations present
on same link may have different

3:13 speeds data link lay provides flow control that allows both sender and receiver to
exchange data at same speed

3:22 the last function is multiaccess when host on the shared link tries to transfer the data
it has a high probability of

3:31 collision data link lab provides mechanisms such as csma CD to equip capability of
accessing a shared media

3:40 among multiple systems next we are going to discuss the first function of data link
layer that is framing the major

3:48 responsibility of data link layer is to make physical link reliable and to do so

3:54 it breaks the network layer data stream into small blocks this process is called
segmentation and in order to form a

4:03 frame it adds header and frame flag to each and every block and this process is

4:10 called encapsulation header of the frame generally contains three Fields the first
field is address field which

4:18 contains the address of sender and receiver the second field is error detecting code
where a check Sum of the

4:27 frame for error detection is stored the Third Field is control field which stores the
additional information to

4:34 implement protocol functions the receiving data link layer must know the start and
the end of a frame according

4:42 to the frame flag a good design must make it easy for a receiver to find the start of
a new frame while using little

4:52 of the channel bandwidth we are now going to look at four of the framing methods
the first method is by count

5:00 this framing method uses a field in the header to describe the number of bytes present in the frame when the receiver's

5:08 data link layer sees this count it comes to know how many bytes to follow the next method is flag bytes

5:16 with bite stuffing in this case each frame starts and ends with special bite sequence generally same bite also called

5:25 a flag bite is used as starting and ending D limiter and this bite sequence is shown in figure as

5:34 flag sometimes it may happen that the flag bite is present in the data then in

5:41 order to solve this problem the sender's data link layer inserts a special bite known as ESC or Escape bite immediately

5:51 before each accidental flag bite in the data this technique is known as bite stuffing the third method is flag bits

6:01 6 minutes, 1 second with bit stuffing as we know that frame flag is a special bit pattern which should not be present any other place

6:08 inside a frame if a special frame flag of 0 6 * 1 0 is used at the start and

6:17 end of the frame and whenever a senders dlc's five consecutive ones in the data then it automatically adds zero after

6:25 five consecutive ones and receiver deletes this Z bit that follows five consecutive one bits in the received data this process is

6:35 called bit stuffing the last method is physical layer coding violations this framing method is used in those networks

6:43 in which the encoding on the physical medium contains some redundancy some lens encode each bit using two physical

6:53 bits and Manchester coding is normally used in this case the one bit is and Ed

6:59 using one0 and 0 bit is encoded using 01 this type of invalid physical code is

7:06 used in 802 lens standards next is flow control modern networks are aimed to support a wide variety of hosts and

7:15 communication mediums suppose a 200 MHz Pentium based host wants to transmit a

7:20 data to 25 MHz based 80386 host naturally the faster Pentium will drone

7:28 the slower 8038 six with data similarly consider Two Hosts and both of them

7:35 using an ethernet land but with the two ethernet connected by a 28.8 kbps model

7:42 link if one host starts transmitting to the other host at ethernet speeds then in this case the model link will quickly

7:51 become overburdened in both of these cases flow control is required to speed the data transfer at suitable rate

8:00 flow control is a technique that informs the sender the amount of data it can transfer before it should

8:07 wait for an acknowledgement from the receiver the flow of data from sender to receiver should not be allowed to

8:14 overburden the receiver receiver must also be able to report to the transmitter before its limits are

8:22 reached and the sender should send less number of frames the limit May well be amount of memory used in order to to

8:29 store the incoming frames or the processing power of receiver the two methods developed for flow control are

8:37 stop and weight and sliding window first we'll discuss stop and weight stop and weight is the simplest form of flow

8:45 control in which a sender transmits a data frame after receiving the data frame the receiver indicates its

8:52 keenness to accept another frame by sending back an acknowledgement frame acknowledging the frame received earlier

9:02 the sender must wait until AK frame is received by it before transmitting the next frame this is also referred to as a

9:11 request reply mechanism which is very easy to understand and implement but not very

9:17 efficient in lens with fast links this is not a major concern but van links will spend most of their time staying

9:25 idle especially if a large number of hops are required figure here shows the working of the

9:32 Stop and waight protocol the blue arrows indicate the sequence of data frames being sent over the link from sender to

9:42 the receiver the protocol depends on two-way transmission either half or full duplex to allow the receiver to return

9:50 acknowledgement frames acknowledging the successful transmission the acknowledgements are shown in green

9:56 color in this diagram which move back to the original sender a small processing delay is introduced between the

10:05 reception of the last data frame and the generation of the acknowledgement major drawbacks of this

10:12 scheme is that only one frame can be transmitted at a time and this leads to inefficiency in the performance if

10:21 propagation delays longer than the transmission delay next we'll calculate the link utilization in stop and waight

10:28 protocol call let's assume the few things the first is transmission time it is the time taken by a station to

10:37 transmit a data frame normalized to a value of one and the second is propagation delay it is a Time taken by

10:45 a bit to travel from sender to receiver expressed as a if the value of a is less
10:52 than one the frame is sufficiently long such that the first bit of the frame arrive at the destination before the
10:59 source has completed transmission of the frame if a is greater than one send a completes transmission of the entire
11:08 frame before the leading bits of the frame arrive at the receiver the link utilization U is equal to $1 / (1 + 2a)$
11:17 where a is equal to propagation time divided by transmission time it is clear from this equation that the link
11:26 utilization is strongly dependent on the ratio of the propagation time to the transmission time when the propagation
11:34 time is very small as it happens in case of lands the link utilization is good but when the propagation delays are very
11:42 long in case of uh Satellite Communication the link utilization can become very poor so to improve the link
11:52 utilization we can use sliding window protocol instead of using stop and weight protocol the next method of flow
12:00 control is sliding window the simple stop and weight protocol does not work well if multiple data frames are used
12:08 for single message only one data frame can be in transmission mode at a time in stop and weight protocol if a is greater
12:17 than one severe inefficiencies result efficiency of the protocol can be improved if multiple frames are
12:24 transmitted at the same time and also by using the full duplex line the sending station keep track on the
12:32 frames by transmitting sequentially numbered frames the sequence number should be of limited size because it
12:40 occupies a field in the data frame if there are K bits in the header of the data frame then the range of the
12:47 sequence numbers must be from 0 to $2^K - 1$ sender keeps a list of sequence
12:55 numbers that it is authorized to send and this list is called a sender window the maximum size of the sender's
13:02 window is $2^K - 1$ the sender is allocated above a space equal to the window size receiver also keeps a window
13:11 called a receiver window of size $2^K - 1$ the receiver acknowledges every frame by sending back
13:18 an acknowledgement frame which includes the sequence number of the expected next frame this also makes a public
13:27 announcement that the receiver is ready to receive the next end frames starting with the specified

13:34 number this scheme is used to acknowledge multiple frames sliding window algorithm provides a flow control

13:42 for network data transfers TCP protocol uses a sliding window protocol a buffer is placed between the application

13:50 program and network data flow by the sliding window protocol for TCP the buffer is normally present in the kernel

13:58 of the operating system the data which is received from the network gets stored in the buffer

14:05 space from where the application can read data at its own speed as the data

14:13 is read by the application buffer space gets empty and is ready to accept more data from the network now we are going

14:21 to discuss the sender and receiver sliding windows at any point of time the sender is allowed to transmit frames

14:29 with sequence numbers in a particular range the center window as shown in the figure here the next is receiver sliding

14:39 window the receiver keeps a window of size one if receiver sees that frame should arrive in a particular order if

14:48 frames are received out of order then it is discarded and needs to be recent however the receiver window increased by

14:57 one if a particular frame is received the frame is acknowledged by sending back an AK frame which includes uh the

15:07 sequence number of the next frame this also openly announces that it is ready to receive the next end frames starting

15:16 with the specified number this scheme can also be used to acknowledge multiple frames sliding window is beneficial if

15:24 local application processes data at the same rate at which it is being transferred if packet size is smaller

15:32 than the window size then multiple packets can be present in the network because the sender knows that the space

15:39 exists on the receiver site to store all the packets ideally steady state can be

15:45 attained where a number of packets in the forward Direction and window announcements in the reverse direction

15:53 are constantly in traveling when the sender receives new window announcement it transmits more data packets and when

16:02 data from the buffer is read by the application more window announcements are generated maintaining a sequence of

16:09 data packets during transfer guarantees the effective use of network resources third function that is error control of

16:18 data link layer when the receiver detects an error in the message or packet it then informs the sender to retransmit the particular message or

16:27 packet the most most popular retransmission scheme is known as automatic repeat request or

16:35 arq the three well-known erq techniques are stop and wait arq go back and erq

16:42 and selective repeat arq first we will discuss stop and weight arq stop and weight arq is the simplest protocol

16:51 among all the protocols in which the sender sends a frame and then it waits till a positive acknowledgement or

16:58 negative acknowledgement is received from the receiver the receiver sends a positive acknowledgement only if the

17:06 frame is received correctly or else it sends back negative acknowledgement the sender sends a new frame only after

17:15 receiving positive acknowledgement otherwise it retransmits the old frame if negative acknowledgement is received

17:23 this is Illustrated in the figure here to deal with the issue of a lost or damag frame the sender is provided with

17:30 the timer if acknowledgement is lost the sender transmits the old frame in the

17:38 figure the second pdu of data is lost during transmission the sender does not know about this loss but starts a timer

17:47 after transmitting each pdu generally a positive acknowledgement pdu is received before the timer

17:54 expires in this case no acknowledgement is received and and the timer counts down to zero and retransmission of the

18:02 same pdu is initiated by the sender the sender always starts a timer immediately after transmitting a pdu but the second

18:10 transmission receives an acknowledgement pdu before the timer expires finally indicating that the data has now been

18:19 received by the remote node as the duplicate frame is received by the receiver therefore it is

18:26 discarded the receiver now can identify that it has received a duplicate frame

18:33 from the label of the frame and it is discarded to address the issue of lost or damaged frames for example a frame

18:41 gets corrupted due to noise during the transmission and there is a concept of negative acknowledgement frames the

18:50 receiver sends back a negative acknowledgement frame to the sender if it finds that the received frame is corrupted when a transmitter receives a

18:59 negative acknowledgement frame before the timeout of the timer the old frame is transmitted again as shown in the

19:06 figure the major advantage of stop and weight erq is its Simplicity it also requires minimum buffer size however it

19:15 makes highly inefficient use of communication links particularly when propagation delay is large

19:22 second arq scheme is go backen arq go backen is one of the most popular arq in which the sender transmits the frames

19:31 continuously without even waiting for acknowledgement that is why this protocol is called as continuous arq

19:39 when the frame is received by the receiver it continues to send acknowledgments or negative acknowledgements back to the sender if

19:47 negative acknowledgement frame is received by the sender it retransmits the frame including all the successive frames as shown in the figure therefore

19:57 this protocol is called go n erq if a frame is lost during transmission then the receiver sends negative

20:05 acknowledgement frame if there is a long delay in sending the N frame the sender will retransmits the Lost frame after

20:14 its timer times out if the AK frame is lost in this case the sender resends the

20:21 frames after its timer times out as shown in the figure suppose a case of full duplex transmission the receiver

20:29 sends back piggybacked acknowledgement by using some number in the acknowledgement field of its data frame

20:37 for example if a 3 bit sequence number is used and suppose that a station sends frame 0er and gets back an rr1 and then

20:47 sends frame 1 2 3 4 5 6 7 and zero and gets another rr1 this might either mean

20:57 that rr1 is a cumulative AK or all eight frames were damaged this uncertainty can

21:05 be defeated if the maximum window size is limited to seven that is for a k bit

21:11 sequence number field it limited to $2^k - 1$ the number n which is equal to 2^k

21:20 minus1 specifies how many frames can be sent without receiving acknowledgement if no acknowledgement is

21:28 received after sending end frames the sender takes the help of a timer after the timer timeout it resumes free trans

21:36 Mission the go backend protocol also takes care of damaged frames and damaged acknowledgements this scheme is a little

21:44 more complex than the previous one but gives much higher throughput the last arq scheme is selective repeat arq the

21:53 selective repeat arq scheme retransmits only those for which n Cas are received

22:00 or for which timer has expired so this is shown in the figure here this is the

22:06 most efficient method among all the arq schemes but the sender must be more

22:13 complex so that it can send out of the order frames the receiver also must have

22:20 storage space to store the post n frames and processing power to reinsert frames in proper sequence so friends this this was all

- 22:28 about data link cair hope all the concepts are clear to you hope to see you in the next lecture till then
- 22:36 goodbye thank you